

Problem 6.1 (MATLAB)

Univariate Multiple Linear Regression (UMLR)

The file '***dataset_problem_6_1.mat***' contains data for the measured characteristics of 76 young bulls (less than two years old) sold at an auction.

The univariate (scalar) response variable, Y , is the bull's height in [inches] at shoulder at the time of the sale.

The two predictor variables, collected in the matrix, Z , is the bull's yearling (one year old) height in [inches] at shoulder and the bull's fat free body weight in [pounds], respectively.

It is desired to build and examine a standard multiple regression model on the form

$$Y = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \varepsilon$$

- Load the data file
- Calculate an unbiased Least Squares estimate of the regression coefficients
- Calculate an unbiased estimate of $\text{Var}[\varepsilon]$
- Perform the SS decomposition: $SST = SSR + SSE$
- Calculate the coefficient of determination, R_sqr , and the adjusted coefficient of determination, R_adj_sqr
- Estimate the covariance matrix for the estimated regression coefficients
- Calculate 95% simultaneous confidence intervals for the regression coefficients
- Calculate 95% Bonferroni confidence intervals for the regression coefficients
- Perform an Extra Sum of Squares based test for significant dependency of Y on (z_1, z_2) (significant regression model), that is test the hypothesis $H_0: \beta_1 = \beta_2 = 0, \beta_0$ arbitrary
- Perform an ANOVA table based test for the same hypothesis and demonstrate that this approach is equivalent to the Extra Sum of Squares approach
- Perform marginal tests $H_{0,i}: \beta_i = 0, i = 0, 1, 2$ and compare the results with above
- Calculate a 95% prediction interval for $E[Y(z_0)]$, the mean of new responses at $z=z_0$, for $z_0 = [z_1 \ z_2] = [50.5 \ 970]$
- Calculate a 95% prediction interval for $Y(z_0)$, a single new response at $z=z_0$
- Perform a model check: analysis of residuals (estimated model errors) by plotting the residuals as function of observation index, response variable and each predictor variable. Also plot a histogram and a QQ-plot for the residuals.
- Conclude on the validity of the model

- Next use the built-in function ***LinearModel.fit*** to create model, estimate regression coefficients, do marginal tests for coefficients, calculate R-squared and adjusted R-squared, test for significant regression model
- Use the built-in function ***anova*** to do marginal tests for coefficients (not intercept)
- Use the built-in method in class LinearModel ***coefCI*** to calculate 95% marginal confidence intervals for coefficients
- Use the built-in method in class LinearModel ***predict*** to calculate 95% confidence intervals for new observations
- Use the built-in methods in class LinearModel ***plotDiagnostics*** and ***plotResiduals*** to perform a model check

Problem 6.2 (Theoretical)

Weighted Least Squares Multiple Linear Regression

Consider the UMLR-model for n observations

$$Y = Z\beta + \varepsilon$$

where as usual

$$E[\varepsilon] = 0,$$

but where the $(n \times n)$ error covariance matrix,

$$\Sigma_{\varepsilon} = \text{Cov}[\varepsilon] = E[\varepsilon\varepsilon^T] = \sigma^2 V$$

not necessarily is proportional to the identity matrix as in the standard Least Squares model.

V can still be diagonal to allow for different variances for each error term, or it can even be non-diagonal (but still symmetrical) to allow for correlation between errors for different observations.

In this case a Weighted Least Squares estimate of regression coefficients is appropriate.

To derive an equation for this estimate

- Multiply the above regression equation with $V^{-1/2}$ and identify a modified regression equation on the form $Y^* = Z^*\beta + \varepsilon^*$.
- Show that $\Sigma_{\varepsilon^*} = \text{Cov}[\varepsilon^*] = E[\varepsilon^* \varepsilon^{*T}] = \sigma^2 I$ indicating that the modified regression equation is on the standard form.
- Give the (standard) Least Squares equation for the estimated regression coefficients, $\hat{\beta}_{LS}$ expressed in Z^* and Y^* .
- Finally use the relation between the two regression equations to find the Weighted Least Squares estimate for the regression coefficients, $\hat{\beta}_{WLS}$ expressed in Z, V and Y .

Problem 6.3 (MATLAB)

Two-factor ANOVA with interactions as a special case of linear regression

In an experiment involving remote sensing, the spectral reflectance at 720 [nm] of three species of 1-year-old seedlings was measured during the growing season.

The matrix **data** in the file '**dataset_problem_6_3.mat**' contains data for these measurements.

The first column in the matrix contains the response variable, the relative spectral reflectance in [%] at 720 [nm].

The second column contains indicators as to whether the specific seedling is of species 1, 2 or 3 (categorical indicator variable = factor 1).

The measurements were taken at three different times during the growing season, and the third column of the matrix contains indicators for the times, coded as 1, 2 or 3 (categorical indicator variable = factor 2).

The setup can thus be interpreted as a two-factor ANOVA model with possible species-time interactions and with 3 levels for factor 1 and 3 levels for factor 2. Since 4 individual seedlings were measured for each species-time combination, a total of $3 \times 3 \times 4 = 36$ measurements are available.

It is desired to formulate and analyze the ANOVA model as a multiple linear regression model (UMLR), $Y = Z\beta + \varepsilon$, checking for any significant factor 1- factor 2 interactions and for any significant factor 1 effect and factor 2 effect.

- Load the data file
- Build response vector Y
- Build the design matrix Z by using (dummy) indicator predictor variables, specifically one for an intercept term in the regression model, three for the levels of factor 1, three for the levels of factor 2 and nine for the possible factor 1- factor 2 combinations, giving a total of 16 columns in Z corresponding to a regression model with 16 regression coefficients.
- Calculate $\text{rank}(Z)$, $\text{rank}(Z^T Z)$ and $\det(Z^T Z)$ and comment. Does $(Z^T Z)^{-1}$ exist ?
- Calculate the pseudo-inverse (generalized inverse), $(Z^T Z)^{-}$
- Calculate the unbiased LS estimate of the regression coefficients using the pseudo-inverse
- Perform an Extra Sums of Squares based test for no significant interactions between species and time, that is test
 H_0 : all regression coefficients corresponding to interaction predictor variables = 0

- Perform an Extra Sums of Squares based test for no factor 1 effect (species), that is test
 H_0 : all regression coefficients corresponding to factor 1 predictor variables = 0
- Perform an Extra Sums of Squares based test for no factor 2 effect (time), that is test
 H_0 : all regression coefficients corresponding to factor 2 predictor variables = 0
- Compare the test results including p-values with an ANOVA2 table calculated with built-in function ***anova2***.